

PhyRC 2027: Physical Robotic Caregiving Challenge: Proposal for an ICRA 2027 Competition

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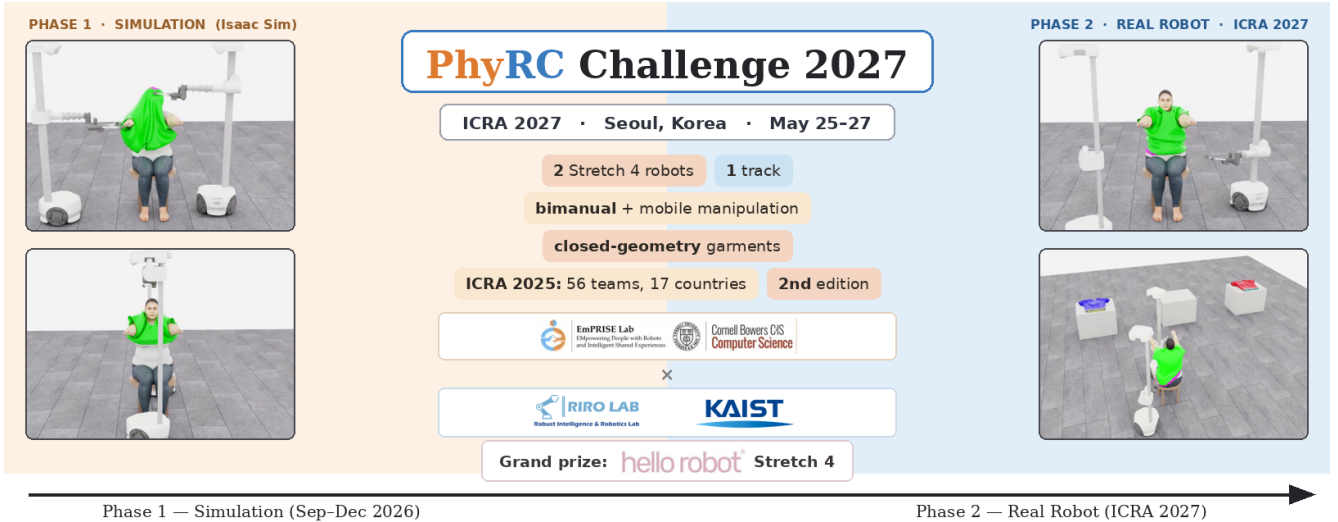


Fig. 1: **PhyRC 2027**: We propose to host the second edition of the **PhyRC (Physical Robotic Caregiving)** Challenge at ICRA 2027 in Seoul, jointly organized by the EmPRISE Lab at Cornell University and the Robust Intelligence & Robotics (RIRO) Lab at KAIST. The competition features a single, deeper task: **bimanual mobile manipulation for robot-assisted dressing** with closed-geometry garments, performed by **two coordinated Hello Robot Stretch-4 mobile manipulators**. **Left**: the Phase 1 environment in NVIDIA Isaac Sim, where the robots thread the garment over the head and route each sleeve (Sep–Dec 2026). **Right**: the completed task, and the wider scene showing the multiple garments staged for randomization. The inaugural PhyRC at ICRA 2025 drew 56 teams from 17 countries and culminated in live real-robot finals in Atlanta. Hello Robot has generously sponsored a Stretch-4 robot as the grand prize. Details are in our website: <https://emprise.cs.cornell.edu/phyrc/challenge-2027/>.

I. WHY THIS COMPETITION AND WHY NOW?

Over one billion people worldwide require assistance to carry out daily tasks, and with a rapidly aging population this number continues to rise [1]. Caregiving robots have the potential to increase care-recipient independence and reduce caregiver workload [2]–[8]. This is felt acutely in the ICRA 2027 host country: Korea is among the fastest-aging societies in the world, making Seoul a particularly fitting venue for a competition on physical robotic caregiving.

In 2025 we ran the **first ever competition for caregiving robots**, the PhyRC Challenge, at ICRA 2025 in Atlanta. It attracted **56 teams from 17 countries** across North Amer-

ica, South America, Europe, Asia, and the Middle East, and culminated in live real-robot finals. That edition answered its central question: a standardized, physically-grounded caregiving benchmark is both feasible and generative of real research. No one was able to solve the complete task. It also exposed the next frontier clearly. Every finalist team in the dressing track was limited by the same structural constraint: a *single, fixed-base arm* cannot simultaneously stabilize a limb and route a garment over it. Human caregivers do not dress people one-handed, and they do not stand still.

PhyRC 2027 is designed around exactly that gap. We propose a single track, robot-assisted dressing with closed-geometry garments, executed by **two coordinated**

Stretch-4 mobile manipulators. Although the physical scene (manikin, chair, garment) is deliberately held constant relative to 2025 so that progress is directly measurable, the embodiment is fundamentally different. Teams must now solve, jointly:

- **Deformable-object manipulation:** closed-geometry garments (t-shirts) have no front opening and must be topologically threaded over the body, unlike the open-geometry hospital gowns used in most prior work (Fig. 2).
- **Bimanual coordination:** two independent robots must share a workspace, hand off implicit responsibility for limb support versus garment routing, and avoid deadlock, all without a shared kinematic base or a single controller.
- **Mobile manipulation:** each Stretch must reposition its base to reach around the body, so navigation and manipulation cannot be decoupled.
- **Physical human–robot interaction:** contact with the care recipient is not incidental but required, and must remain within safe force limits throughout.

To the best of our knowledge, no existing benchmark or competition couples all four.

Real-world competitions are a proven way to measure and accelerate progress in emerging fields. The Amazon Picking Challenge [9] drove significant advances in robotic manipulation, and the DARPA Robotics Challenge [10] played an important role in developing robots capable of assisting humans in disaster scenarios. PhyRC 2025 demonstrated that this model transfers to caregiving. PhyRC 2027 deepens it.

See also our website at <https://emprise.cs.cornell.edu/phyrc/challenge-2027/>.

II. COMPETITION SUMMARY

PhyRC 2027 has a **single track**:

- **Track:** Bimanual Mobile Manipulation for Robot-Assisted Dressing.

The competition runs in two phases:

- **Phase 1:** Simulation (Sep 15 – Dec 15, 2026), in NVIDIA Isaac Sim.
- **Phase 2:** Real Robot (Jan – May 2027), culminating in live finals **proposed for ICRA 2027**.

Top teams from Phase 1 are invited to Phase 2, prepare remotely from January, and compete on-site in Seoul. Table I summarizes what changes relative to the inaugural edition.

a) *Why one track:* Two tracks in 2025 diluted both the participant pool and our own organizing capacity, and required two separate arenas. Concentrating on dressing lets us raise task difficulty substantially, halve our footprint request, run a tighter three-day on-site schedule, and produce a cleaner year-over-year comparison against the 2025 results.

b) *Why Isaac Sim:* RCareWorld served the 2025 edition well, but Isaac Sim offers GPU-parallel simulation for large-scale policy training, a far larger existing user base, and native support for the Stretch platform. Together these lower the barrier for teams who have never worked in caregiving before.

TABLE I: PhyRC 2025 (ICRA 2025) vs. PhyRC 2027 (proposed).

	PhyRC 2025	PhyRC 2027
Tracks	2 (dressing, bed bathing)	1 (dressing)
Embodiment	Single fixed-base arm	Two Stretch-4 mobile manipulators
Core skills	Deformable manip. + pHRI	+ bimanual coordination + mobile manip.
Simulator	RCareWorld (Unity)	NVIDIA Isaac Sim
Organizers	Cornell	Cornell × KAIST
Prize	Kinova Gen 3, Stretch 3	Stretch-4
Venue	Atlanta, USA	Seoul, Korea

III. SPONSORSHIP, FUNDING, AND BUDGET

A. Funding Sources

- **Hello Robot** sponsors the grand prize (a Stretch-4 robot), provides **two Stretch-4 robots on loan at the venue** for teams to compete with, assists with on-site setup, and, critically for equitable access, provides **remote access to Stretch hardware** so that Phase 2 teams can develop and practice on real robots before traveling to ICRA.
- **KAIST** provides local equipment, task hardware (manikins, chairs, arm rests, garments), on-site personnel, and local logistics support as the host-country organizer.
- **Shipping support** for participating teams: For shipping two Stretch-4 robots from Hello Robot, KAIST provides **USD 10,000** in support through the *Samsung Electronics Robotics Talent Development Program (Research)* (Project Code: A0602030010).
- Teams may also bring their own robots at their own cost.

B. Estimated Budget

Table II gives our estimate. The large majority is covered in kind by sponsors and the host institution. We are not requesting funds from the Organizing Committee beyond the facility support in Section IV.

C. Awards

- **Grand Prize:** a Hello Robot Stretch-4 robot to the winning team.
- **Runner-up and third place:** certificates and public recognition at the competition and on the challenge website.
- **Best Innovation Award:** recognizing the most novel technical approach, judged by the organizing committee, independent of final score.

IV. REQUESTED FACILITY SUPPORT

Since we have consolidated from two tracks to one, our footprint request is roughly **half** that of PhyRC 2025. Fig. 4 shows the proposed layout.

- **Arena size:** an arena of at least **6 m × 5 m (30 m²)**. Two mobile bases must circulate around a seated manikin, so the space around the chair matters more than for a fixed arm.

TABLE II: Estimated budget (USD). Figures are planning estimates.

Item	Source	Est.
Grand prize: Stretch-4 robot	Hello Robot	30,000
2 loaner Stretch-4 robots at venue	Hello Robot	in kind
Remote Stretch access (Phase 2 prep)	Hello Robot	in kind
On-site setup and technical support	Hello Robot	in kind
Manikins (2 $\times$, incl. backup)	KAIST	3,000
Chair, arm rests, garments, consumables	KAIST	1,500
Arena fencing, signage, printing	KAIST	1,000
Local equipment transport to venue	KAIST	2,000
Cloud/GPU for Phase 1 auto-evaluation	Organizers	2,500
Livestream production (Sec. VII)	Organizers	1,500
Robot shipping, Hello Robot $\rightarrow$ Seoul	KAIST (Samsung)	10,000
Total (excl. in-kind)		$\sim$51,500

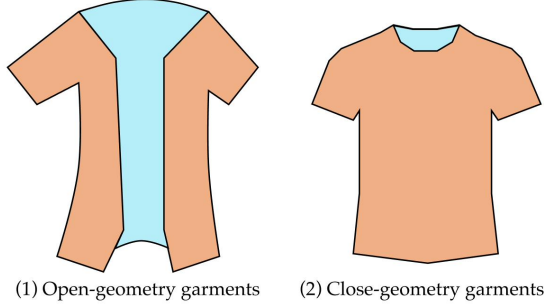


Fig. 2: Open-geometry vs. closed-geometry garments: open-geometry garments such as hospital gowns and jackets have openings on the front or back, making dressing substantially easier. PhyRC 2027 focuses exclusively on closed-geometry garments such as t-shirts, which must be topologically threaded over the body.

- **Staging area:** approximately $5\text{ m} \times 4\text{ m}$ (20 m^2) adjacent to the arena for teams to stage robots, charge batteries, and work between runs.
- **Total:** approximately 50 m^2 . We are happy to negotiate if the venue has hard space constraints.
- **Minimum ceiling height:** 3 m . The Stretch telescoping mast plus overhead camera mounting requires vertical clearance.
- **Power:** at least 6 standard outlets (Korean Type-C/F, 220 V) within the arena and staging area, plus extension cords and power strips. Robot charging runs continuously during the event.
- **Networking:** a wired network jack in the arena. We will bring our own router to establish a dedicated local WiFi environment, as competition-grade teleoperation and

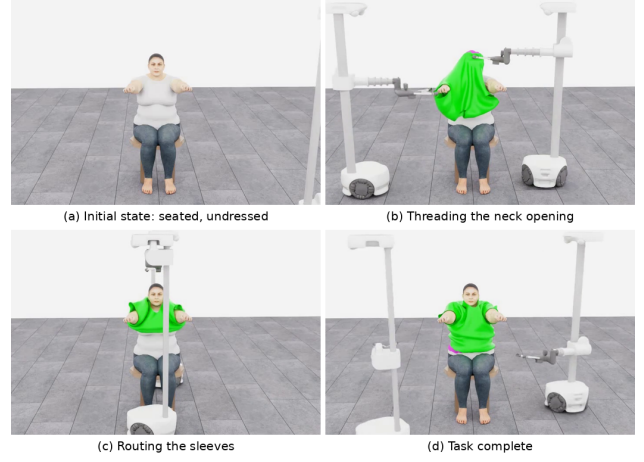


Fig. 3: **Phase 1 simulation environment** in NVIDIA Isaac Sim. Two Stretch-4 mobile manipulators coordinate to dress a seated human avatar in a closed-geometry t-shirt. (a) The avatar begins seated and undressed. (b) Both robots lift the garment and thread the neck opening over the head, the step that has no analogue in open-geometry dressing. (c) The robots reposition their bases and route each sleeve over an arm. (d) Task complete. Note that at no point can a single arm accomplish the step alone: one robot maintains garment tension while the other advances the fabric.

logging cannot rely on shared conference WiFi.

- **Furniture:** 4 tables and 10 chairs for the staging area.
- **Audiovisual:** one large monitor or projector at the arena for live scoring and for teams' final presentations. If a nearby room with a projector is available for the closing presentations, that would be ideal.
- **Utilities:** tape, white paper, and access to an on-site printing service if available.
- **Advertisement:** social media promotion from the conference channels would be appreciated (optional).

V. COMPETITION TIMELINE

- **September 1, 2026:** Registration opens.
- **Oct 15, 2026:** Registration deadline (11:59pm AoE).
- **Sep 15 – Dec 15, 2026: Phase 1 (Simulation)**, 3 months. Teams develop and submit policies against the Isaac Sim environment, and submissions are auto-evaluated on a leaderboard.
- **Dec 22, 2026:** Phase 1 results announced, and top teams invited to Phase 2.
- **Jan – May 2027: Phase 2 preparation.** Teams develop on real hardware, using Hello Robot's remote Stretch access and/or their own robots. Organizers hold regular online office hours.
- **May 25 – 27, 2027:** On-site finals at ICRA 2027, Seoul.

A. Three-Day On-Site Schedule

The following schedule fits the May 25–27 competition window and concludes **before the award ceremony**. Since

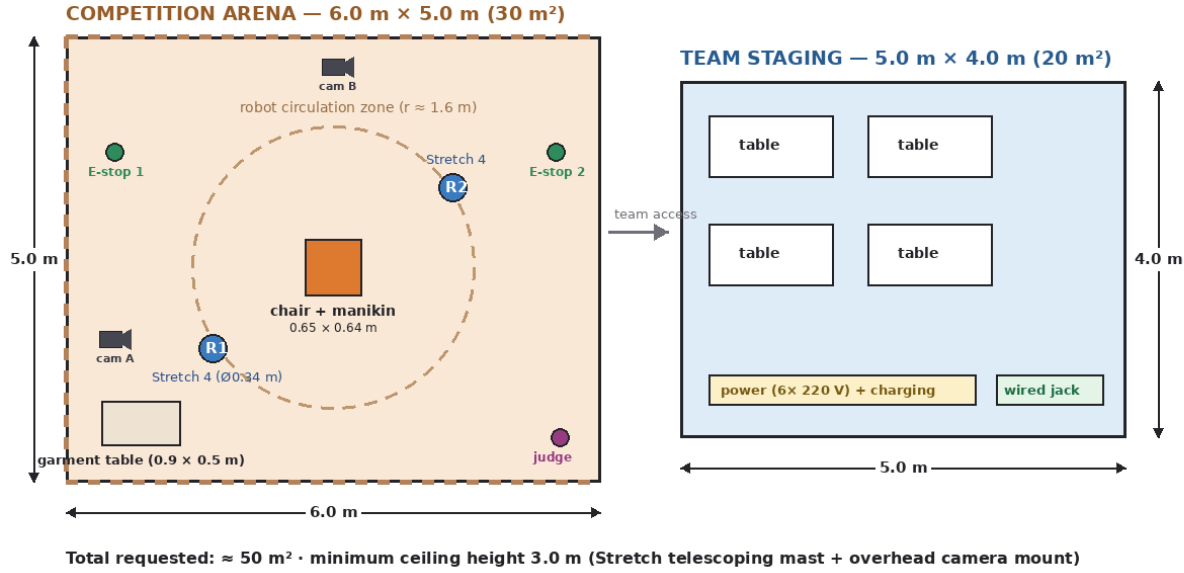


Fig. 4: **Proposed arena layout and space request.** The competition arena (left) is sized for two mobile bases to circulate freely around the seated manikin. The binding constraint is not the manikin footprint but the clearance required for a robot to drive around to the opposite shoulder. Camera positions for the final-round broadcast (Sec. VII) and the two mandatory e-stop operator stations are indicated. The adjacent staging area (right) holds team tables, robot charging, and the wired network drop. Total request is approximately 50 m² with a minimum ceiling height of 3.0 m.

two Stretch-4 robots are shared across teams, practice and competition slots are explicitly allocated.

a) Day 1, Monday, May 25: Setup and Practice:

- 08:00–10:00: Organizer arena setup, robot unpacking, calibration, and network configuration.
- 10:00–11:00: Team check-in, safety briefing, rules walk-through, Q&A.
- 11:00–18:00: Rotating practice slots (approx. 75 min per team) on the competition setup.

b) Day 2, Tuesday, May 26: Qualification Runs:

- 08:30–09:00: Setup verification and randomization seed draw.
- 09:00–12:30: Qualification round, part 1 (teams in drawn order).
- 12:30–13:30: Lunch, with the arena open for public viewing and demonstrations.
- 13:30–17:00: Qualification round, part 2.
- 17:00–18:00: Scores posted, followed by a short practice window for finalists.

c) Day 3, Wednesday, May 27: Finals:

- 08:30–09:00: Final setup and briefing.
- 09:00–12:00: **Final round** (livestreamed, see Sec. VIII). Each finalist performs scored trials before judges and audience.
- 12:00–12:45: Judging, score reconciliation, Best Innovation deliberation.

- 12:45–13:45: Team technical presentations (5–7 min each).
- 13:45–14:15: **Winners announced**, prize presentation, and group photo.
- 14:15–17:00: Teardown and shipping preparation.

This leaves the remainder of May 27 and all of May 28 clear of competition activity, so results are final well ahead of the conference award ceremony.

VI. EXPECTED PARTICIPATION

A. Estimated Numbers

Based directly on PhyRC 2025 (56 registered teams from 17 countries, of which 5 advanced to the real-robot dressing finals) and adjusting for a single track (which concentrates rather than splits registrations), the move to Isaac Sim (a much larger user base), and an Asian venue (better access for teams across Asia), we estimate:

- **Phase 1:** 60–80 registered teams from 15–20 countries, and roughly 250–350 individual participants. We expect 25–35 teams to submit at least one valid evaluated solution.
- **Phase 2:** 5–6 teams invited to the on-site finals.
- **On-site:** approximately **25–30 participants** (up to 5 per team), plus 7 organizers and Hello Robot technical staff.
- **Spectators:** PhyRC 2025 attracted sustained conference-attendee traffic throughout the event, and we anticipate similar or higher interest in Seoul.

B. Possible Participating Teams

The following teams reached the real-robot finals of PhyRC 2025 and are natural candidates to return:

- **UNICHAMPions**: UNICAMP / H.IAAC, Brazil (*winner*, dressing track 2025)
- **RoboNotts**: CHART, University of Nottingham, UK (*winner*, bed bathing track 2025)
- **USC SLURM**: University of Southern California, USA
- **RALLA**: University of York, UK
- **UWMTR**: University of Waterloo, Canada
- **EDLAB**: Shanghai Jiao Tong University, China
- **Idiap**: Idiap Research Institute, Switzerland

In addition, the KAIST co-organizers will actively recruit teams from the Korean and broader East Asian robotics community, including KAIST, SNU, POSTECH, and Korean robotics industry labs. This is a constituency underrepresented in the 2025 edition and well positioned to participate given the Seoul venue. Hello Robot’s user community (several hundred labs worldwide operating Stretch robots) forms a further natural recruitment pool, because the competition platform is one they already own.

C. Outreach and Advertising

We will promote the competition through the following channels:

- robotics mailing lists (robotics-worldwide, HRI, ML-news, compcop),
- the organizers’ direct academic networks at Cornell and KAIST,
- Hello Robot’s social media and community newsletter channels,
- the PIs’ personal social media channels,
- the retained mailing list of all PhyRC 2025 registrants.

VII. BROADCASTING PLAN

The final round on Day 3 will be produced for both on-site and remote audiences:

- **Live stream**: the final round will be livestreamed on the EmPRISE Lab YouTube channel, with the link publicized in advance to the challenge mailing list and via the conference channels.
- **Camera setup**: we will operate a minimum of three cameras: a wide arena shot, a close-up on the manikin and garment (where the decisive task detail occurs), and a roaming camera, all switched through a simple live production setup we will bring ourselves.
- **Live scoring overlay**: the arena monitor will display the running scoreboard, which we will also composite into the stream so remote viewers can follow the competition.
- **Commentary**: an organizer will provide live commentary explaining task structure and scoring to a general audience, which was well-received in 2025.

- **Recording and archive**: all final-round runs will be recorded and published afterwards on the challenge website as a public record and teaching resource, alongside teams’ technical presentations.
- **Media**: we will coordinate with the ICRA media team and with Hello Robot’s communications staff for coverage, and will make organizers available for interviews.

We request no broadcasting equipment from the Organizing Committee beyond the network jack and arena monitor listed in Section IV. If the venue offers a house AV feed, we would gladly integrate with it.

VIII. COMPETITION RULES

Full guidelines will be distributed to participants as Phase 1 concludes. In draft:

- 1) The reference platform is **two Hello Robot Stretch-4 mobile manipulators**. Two units will be available at the venue on loan from Hello Robot and shared across teams on a scheduled rotation.
- 2) Teams may alternatively bring their own robots at their own cost, provided the system consists of two independently mobile manipulators of comparable scale. Teams intending to do so must notify organizers by Apr 25 so the arena and schedule can accommodate them.
- 3) There are no restrictions on algorithms, control approaches, additional sensors, or third-party libraries.
- 4) Each on-site team may have no more than 5 members present in the arena.
- 5) Each team must designate one member responsible for holding the emergency stop whenever the robots are in motion. Teams unable to provide this person must notify organizers in advance, and we will assign a volunteer.
- 6) A spotter from the organizing team will be present to record scores for every scored run.
- 7) Competition trials will be limited to a time-limit that we will finalize based on the overall schedule. Scores are not recorded after the time limit expires. Teams may retry as many times as they wish within the limit, and the highest-scoring attempt counts.
- 8) In the event of technical failure, teams are given a limited contingency window to troubleshoot or switch to backup equipment if available. No additional time is granted beyond this window.
- 9) Conference and competition organizers are not liable for damage to participants’ robots, sensors, or other hardware.
- 10) Teams are responsible for ensuring their robots meet safety standards. Organizers reserve the right to halt any activity if safety concerns arise.
- 11) Finalists are required to give a short technical presentation at the conclusion of the competition.

IX. COMPETITION TASK DETAILS

The physical scene is deliberately held constant relative to PhyRC 2025 so that improvement is directly measurable across editions. What changes is the robot embodiment and

therefore the space of viable solutions. The task is performed on a manikin, so **no humans are involved** as task subjects. This avoids IRB, safety, and privacy complications entirely. All task hardware is provided by the organizers at the venue, and all specifications, dimensions, and purchase links are published in advance so teams can replicate the setup for practice.

A. Motivation

In robot-assisted dressing, a robot helps a care recipient put on clothing. Prior work has largely studied dressing with arms held in the air [5], [6], [11], a pose that care recipients with mobility limitations cannot sustain, and has largely used open-geometry garments (hospital gowns, jackets) that open at the front (Fig. 2). PhyRC 2025 established closed-geometry garments and naturally hanging arms as a tractable but hard benchmark. PhyRC 2027 keeps that benchmark and removes the single-arm crutch: with two mobile manipulators, one robot can support or reposition a limb while the other routes the garment, mirroring how human caregivers actually work. It also requires the team to solve coordination, workspace sharing, and mobile base placement.

B. Task Description

Two Stretch-4 robots must dress a seated manikin in a closed-geometry garment. We use the **Simulaid Flexible Rescue Randy** manikin¹ seated in a chair without armrests, and an **oversized white short-sleeve v-neck t-shirt**². As in 2025, two optional **arm rests** are available to temporarily support the manikin’s arms. We release CAD files and assembly instructions, and provide both 3D-printed and off-the-shelf options at the venue. Full manikin anthropometry, joint randomization ranges, and garment dimensions are given in Fig. 5. Note that with two robots, teams may now choose to have one robot perform the supporting role the arm rest previously played, an explicit strategic trade-off we expect to differentiate approaches.

C. Initial State

Robots begin with the t-shirt grasped (any part of the garment may be grasped, by either or both robots). Teams may place both mobile bases anywhere in the arena, provided that neither robot nor the garment is in contact with the manikin at the start. If arm rests are used, the manikin’s hands rest on them. Otherwise the arms hang down naturally.

D. Task Variation

- **Garment initial state:** the garment’s initial configuration on the staging surface is randomized.
- **Garment instance:** multiple closed-geometry garments of differing colors are staged in the environment (Fig. 3), so policies cannot overfit to a single cloth model.
- **Body pose:** the shoulder joint is randomized within $[-20^\circ, 20^\circ]$ and the elbow joint within $[-45^\circ, 45^\circ]$.

¹Simulaid Flexible Rescue Randy Manikin

²Amazon Essentials Women’s Short-Sleeve V-Neck T-Shirt, size 2X.

Randomization seeds are drawn publicly at the start of each competition day and are not disclosed to teams in advance.

E. Assessment and Scoring

a) *Phase 1 (Simulation):* Submissions are automatically evaluated in Isaac Sim over multiple randomized initial states. Teams are scored on:

- Pick up the garment (5 pts): the garment remains in a robot’s gripper for 3 seconds while lifted clear.
- Put one wrist in one sleeve (5 pts).
- Pull the garment to the other side (5 pts): any part of the garment passes beyond the opposite shoulder joint.
- Dress the other sleeve (5 pts).
- Overall dressing (30 pts): $(n\%) \cdot 10 + (m\%) \cdot 20$, where n is the fraction of the first arm covered and m the fraction of the second.

Efficiency is accounted for by dividing the score by total task time. Teams may submit multiple times per day, and the highest score counts.

b) *Phase 2 (Real Robot):* Scored on the same task structure, measured physically:

- **Put one wrist in one sleeve** (5 pts).
- **Pull the t-shirt to the other side** (3 pts): any part passes beyond the opposite shoulder joint.
- **Neck dressing** (5 pts): the neck is through the neck hole, regardless of v-neck orientation.
- **Correct facing** (3 pts): the v-neck ends up on the front of the body.
- **Dress the other sleeve** (5 pts).
- **Overall dressing** (30 pts): $(n\%) \cdot 10 + (m\%) \cdot 20$, measured manually as above.
- **Coordination bonus** (up to 10 pts): awarded for demonstrated productive use of both robots, for example one robot stabilizing a limb while the other routes the garment.
- **Safety penalty:** excessive force or unsafe contact with the manikin incurs point deductions. Repeated violations end the run.

Ties are broken by total time taken.

X. POTENTIAL RISKS AND MITIGATION STRATEGIES

The safety of participants and attendees is our top priority. Note that PhyRC 2025 ran its full real-robot finals at ICRA 2025 with no safety incidents, and we carry those protocols forward with additions specific to the two-robot setting.

- **Use of a manikin:** no live human subjects are involved at any point, eliminating injury, privacy, and IRB concerns associated with physical human–robot contact.
- **Emergency stop protocol:** every robot is equipped with an emergency stop, and a designated person holds it whenever robots are in motion. With two robots operating simultaneously, we require **one e-stop holder per robot**.
- **Multi-robot collision safety:** because two mobile bases share a confined workspace, teams must demonstrate collision-avoidance behavior during Day 1 practice before

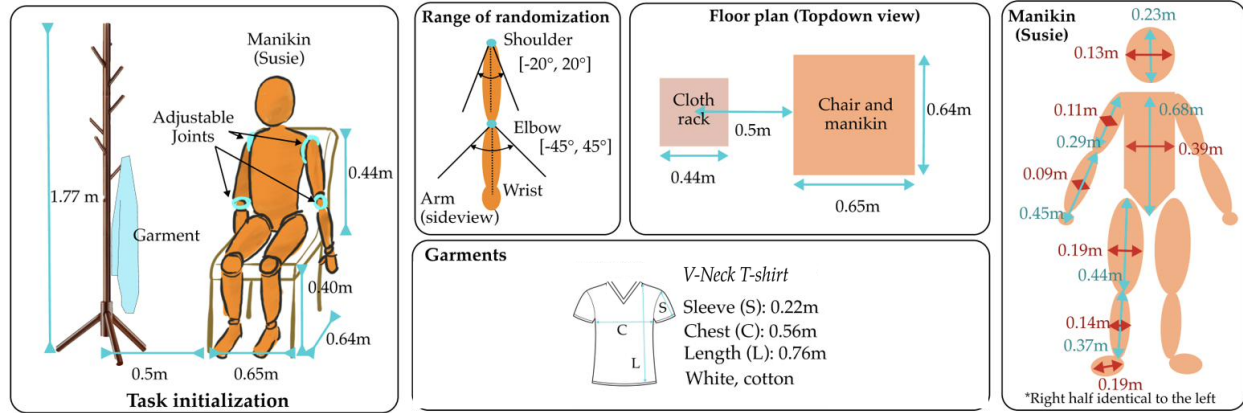


Fig. 5: **Physical task specification** The manikin anthropometry (right), the joint randomization ranges (centre-left), and the garment dimensions (centre-bottom) for 2027, so that results are directly comparable across editions and so that teams can replicate the setup for practice from published specifications. Two elements differ in 2027: the *floor plan* panel shown here was sized for a single fixed-base arm and is superseded by Fig. 4, and the garment set is restricted to closed-geometry items.

being cleared for scored runs. Force and velocity limits are enforced in the arena.

- **Secured competition area:** the arena is fenced off, with entry restricted to authorized participants and organizers during runs.
- **Clear signage:** restricted zones and safety warnings are posted around the arena.
- **Backup equipment:** we provide a backup manikin, backup garments, and spare arm rests. Hello Robot’s on-site presence provides technical support for the loaner robots.
- **Battery and charging safety:** robot charging is confined to the staging area, supervised, and away from public circulation.

XI. HISTORY OF THE COMPETITION

A. PhyRC 2025 at ICRA 2025

The inaugural PhyRC Challenge was the first competition of its kind for caregiving robots. Phase 1 ran Aug–Dec 2024 in RCareWorld, drawing **56 teams from 17 countries** (Fig. 6), evaluated automatically via EvalAI. Five teams advanced in the dressing track and four in bed bathing. The real-robot finals were held at ICRA 2025 in Atlanta, where **UNICHAMPions** (UNICAMP / H.IAAC, Brazil) won the dressing track and received a Kinova Gen 3 7-DoF arm, and **RoboNotts** (University of Nottingham, UK) won the bed bathing track and received a Hello Robot Stretch 3.

The 2025 edition demonstrated that the format works: teams from four continents built working caregiving systems, transferred them from simulation to hardware, and competed publicly. It also demonstrated the operational maturity of the organizing team. Shipping, arena setup, scoring, and safety were all executed without incident. The closest related event remains Cybathlon [12], which focuses

on assistive devices used by people with disabilities rather than caregiving robots. We share its mission but not its scope.

B. Lessons Carried Into 2027

- **Hardware access was the main barrier to entry** between phases. Hello Robot’s remote-robot access for Phase 2 teams directly addresses this.
- **Two tracks stretched organizer capacity thin.** One track, done deeper.
- **Sim-to-real gap was the dominant failure mode.** Isaac Sim plus a shared reference platform (Stretch-4 in both phases) narrows it considerably. In 2025, teams simulated one robot and competed with whatever hardware they owned.
- **Office hours and a discussion forum mattered.** We will run both again from the start of Phase 1.

In addition, we have developed a **teleoperation interface** for the dual-Stretch setup, which lets participants drive both robots directly to collect demonstration data and to build intuition for the coordination problem before writing autonomous policies.

C. RCareWorld Hackathon at Cornell

In preparation for the first PhyRC competition, we hosted an RCareWorld Hackathon at Cornell in April 2024³ with Cornell Big Red APDI, a student organization focused on assistive technology. 96 teams registered, spanning freshmen through seniors across Computer Science, Mechanical Engineering, Operations Research, Biology, Information Science, and other majors (Fig. 7). One winning team’s teleoperation interface⁴, using iOS devices as a space mouse, was subsequently integrated into RCareWorld.

³<https://emprise.cs.cornell.edu/hackathon/>

⁴RCRemote Interface



Fig. 6: Global participation in PhyRC 2025 Phase 1: 56 teams from 17 countries or regions across North America, South America, Europe, Asia, and the Middle East.

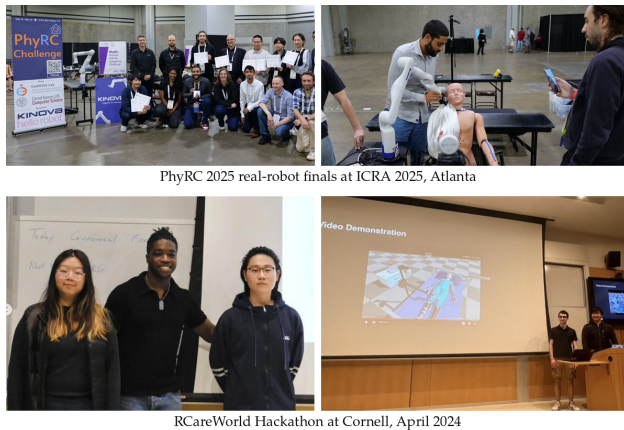


Fig. 7: **Top:** the PhyRC 2025 real-robot finals at ICRA 2025 in Atlanta, showing finalist teams with their certificates alongside the host and sponsor banners, and a competing team dressing the manikin with a Kinova Gen 3 arm under judging. **Bottom:** the RCareWorld Hackathon held at Cornell in April 2024, where 96 student teams competed in the simulator that preceded the challenge, showing a winning team with the organizers and a team presenting its simulated bed bathing solution.

XII. ORGANIZERS

Short bios, contact information, and relevant organizing experience follow. [All the complete CVs are here.](#)

- **Sehee Jung:** Sehee Jung is a postdoctoral researcher in the 5D AI Robotics Initiative at KAIST, working with Dr. Daehyung Park. He received his Ph.D. in Industrial Engineering (minor in Computer Engineering) from North Carolina State University, where he developed computer vision-based biomechanical methods for workplace safety (advisor: Xu Xu). His work received an Honorable Mention at HFES'25. He previously worked as a research engineer at DeepVisions, developing computer vision models for mobile and embedded systems, work that received a CES 2024 Innovation Award. His current research applies computer vision and planning to robotic manipulation and humanoid navigation. He leads the development of the simulation environment and evaluation pipeline for Phase 1, and the real-robot task setup and on-site logistics for the Phase 2 competition in Seoul.

Contact: sjung9@kaist.ac.kr

Relevant events organized:

- PhyRC Challenge 2027, Phase 1 (Simulation), 2026

- **Injin Choi:** Injin Choi is a researcher in the Department

of AI Computing at KAIST. He contributes to the simulation environment development and the Phase 1 evaluation infrastructure for PhyRC 2027.

Contact: changing@kaist.ac.kr

Relevant events organized:

- PhyRC Challenge 2027, Phase 1 (Simulation), 2026

- **Seoyul Jeon:** Seoyul Jeon is a researcher in the School of Computing at KAIST, in the RIRO Lab. He contributes to the real-robot task setup and on-site logistics for the Phase 2 competition in Seoul.

Contact: jsy628496@kaist.ac.kr

Relevant events organized:

- PhyRC Challenge 2027, Phase 1 (Simulation), 2026

- **Qiandao Liu:** Qiandao Liu is a researcher in Computer Science at Cornell University and a member of the EmPRISE Lab. His research lies at the intersection of robotics, learning, and control, with a focus on developing adaptive and generalizable robotic systems. For PhyRC 2027, he contributes to competition task design, participant support, and continuity with the 2025 edition.

Contact: ql342@cornell.edu

Relevant events organized:

- [PhyRC Challenge 2025](#) - RCareWorld technical support

- [PhyRC Challenge 2027](#) - Competition website development and deployment

- **Binit Shah:** Binit Shah is the Senior Software Lead and Developer Relations at Hello Robot, where he leads the software architecture for Stretch. He maintains and releases nearly all of Hello Robot's software stack as open-source, supporting its adoption across the research and developer communities. He received his B.S. and M.S. in Computer Science from the Georgia Institute of Technology. For PhyRC 2027, he contributes to hardware platform support and real-robot integration for the Phase 2 competition.

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Relevant events organized:

- [MoveIt Workshop](#) - ROS World 2021
- Navigation University – ROSCon 2023
- [PhyRC 2025 as Sponsor](#) - ICRA 2025
- [Navigation University](#) - ROSCon 2026

- **Daehyung Park:** Daehyung Park is an Associate Professor at the School of Computing, KAIST, Korea, leading the RIRO Lab. His research lies at the intersection of mobile manipulation, artificial intelligence, and human-robot interaction to advance collaborative robot technologies. Prior to joining KAIST, he had been a postdoctoral associate in Computer Science and Artificial Intelligence Laboratory (CSAIL) at Massachusetts Institute of Technology (MIT), working on inverse constraint learning and semantic knowledge estimation. He received a Ph.D. in Robotics at Georgia Institute of

Technology, an M.S. from the University of Southern California, and a B.S. from Osaka University. Prior to joining his Ph.D., he served as a Robotics Researcher at Samsung Electronics Inc from 2008-2012. His work was awarded the Outstanding Planning Paper Award at ICRA 2023, Google Research Scholar Award 2022, an Outstanding Navigation Award Finalist at ICRA 2022, etc. See more at: <https://rirolab.kaist.ac.kr/>

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Relevant events organized:

- Humanoids 2024 Workshop on [Designing Interactive Humanoids: Learning Tasks through Interaction with Humans](#)

- RSS 2023 Workshop on [Experiment-oriented Locomotion and Manipulation Research](#)

- RO-MAN 2023 Special Session on Human-Agent/Robot Interaction in Healthcare and Medicine

- **Tapomayukh “Tapo” Bhattacharjee:** Tapo is an Associate Professor in the Department of Computer Science at Cornell University, where he directs the EmPRISE Lab (emprise.cs.cornell.edu). He completed his Ph.D. in Robotics at the Georgia Institute of Technology and was an NIH Ruth L. Kirschstein NRSA postdoctoral research associate in Computer Science & Engineering at the University of Washington. His work spans human-robot interaction, haptic perception, and robot manipulation, focused on leveraging robot-world physical interactions in unstructured human environments to perform activities of daily living. He is a recipient of the TRI Young Faculty Researcher Award '24 and the NSF CAREER Award '23. His work has won Best Systems Paper Award at HRI'26, Best Paper Award at RSS'25, Best Paper and Student Paper Award Finalist and Best HRI Paper Award Finalist at ICRA'25, Best Systems Paper Award Finalist at HRI'24, Best Demo Award at HRI'24, Best RoboCup Paper Award at IROS'22, Best Paper Award Finalist and ABB Best Student Paper Award Finalist at IROS'22, Best Technical Advances Paper Award at HRI'19, and Best Demonstration Award at NeurIPS'18. His work has also been featured in many media outlets including the BBC, Reuters, New York Times, IEEE Spectrum, and GeekWire and his robot-assisted feeding work was selected to be one of the best interactive designs of 2019 by Fast Company. He led the organization of PhyRC 2025.

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Relevant events organized:

- [PhyRC Challenge 2025](#) at ICRA 2025 (both phases)
- [RCareWorld Hackathon](#) at Cornell, 2024
- HRI 2024 Workshop on [Assistive Applications, Accessibility, and Disability Ethics](#)
- IROS 2021 [Workshop on RoboTac21](#)
- ICRA 2021 Workshop on [Learning For Caregiving Robots](#)

- IROS 2019 Workshop on [RoboTac 2019](#)
- Humanoids 2019 Workshop on [Robotic Food Manipulation](#)
- IJCAI 2019 Workshop on [Artificial Intelligence and Food](#)
- RSS 2017 Workshop on [Tactile Sensing for Manipulation](#)

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